Next.js Detailed Revision Notes (5-10 Pages)

1. Introduction

Next.js is a React framework that supports Server-Side Rendering (SSR), Static Site Generation (SSG), Incremental Static Regeneration (ISR), API routes, image optimization, routing, and full-stack development. It improves SEO, performance, and developer experience.

Example: Use practical examples in projects such as dashboards, e-commerce, blogs, authentication systems, and admin panels to understand where this feature fits.

2. Project Structure

The App Router uses the app/ folder. page.tsx creates routes, layout.tsx provides shared layouts, loading.tsx shows loading UI, error.tsx handles errors, not-found.tsx customizes 404 pages, route.ts creates API endpoints, globals.css stores global styles.

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3. Server vs Client Components

Server Components render on the server by default and can directly fetch data, reducing bundle size. Client Components require 'use client' and are used for state, event handlers, browser APIs, useEffect, and animations.

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4. Routing

Folders become routes automatically. Dynamic routes use [id]. Catch-all routes use [...slug]. Optional catch-all uses [[...slug]]. Use Link from next/link for navigation and useRouter(), usePathname(), and useSearchParams() from next/navigation.

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5. Data Fetching

Server Components can be async. fetch() is extended by Next.js. cache:'force-cache' caches responses. cache:'no-store' disables cache. next:{revalidate:60} enables ISR by regenerating pages periodically.

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6. Rendering

SSR renders on every request. SSG renders during build time. ISR combines static generation with periodic updates. CSR renders entirely in the browser using Client Components.

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7. Server Actions

Server Actions allow forms and mutations without traditional API routes. Mark functions with 'use server'. Useful for database inserts, updates, authentication, and form handling.

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8. API Routes

Create app/api/users/route.ts. Export GET, POST, PUT, DELETE, PATCH handlers. Return Response.json(). API routes are useful for backend logic, webhooks, and external integrations.

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9. Metadata & SEO

Use export const metadata or generateMetadata(). Configure title, description, Open Graph, Twitter cards, robots, icons, and canonical URLs to improve SEO.

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10. Images & Fonts

next/image automatically optimizes images with lazy loading and resizing. next/font downloads Google fonts at build time, improving Core Web Vitals.

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11. Middleware

middleware.ts runs before requests. Common uses include authentication, redirects, rewrites, localization, and protecting routes.

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12. Environment Variables

Store secrets in .env.local. Variables prefixed with NEXT_PUBLIC_ are exposed to the browser. Others remain server-only.

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13. Deployment

Next.js is optimized for Vercel but can run on Node.js servers. Use npm run build then npm start. Edge Runtime can improve latency for certain workloads.

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14. Interview Revision

Know differences between SSR, CSR, SSG, ISR; App Router vs Pages Router; Server vs Client Components; caching; revalidation; API routes; middleware; server actions; image optimization; metadata; dynamic routing.

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15. Cheat Sheet

page.tsx = route layout.tsx = shared layout loading.tsx = loading UI error.tsx = error boundary route.ts = API endpoint 'use client' = client component 'use server' = server action next/link = navigation next/image = optimized images revalidate = ISR

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